

Analysis group meeting

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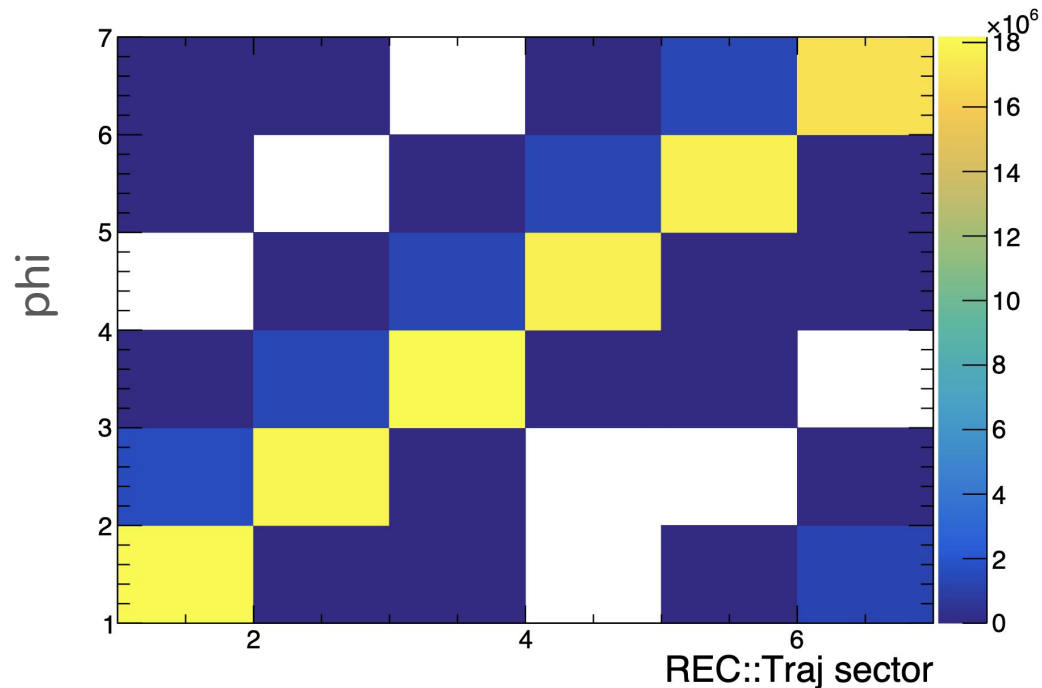


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- Fix some issues in COATJAVA report by Raffaella and Nathan:
 - Issue with the loading of the model on the farm due to the download of a dll
 - Move the AI model from reconstruction/alert to etc/nnet, and change the way to load it.
 - Create a new class to load the model and use it (Nathan's recommendation)
 - Update the wires checked during pre-clustering to ensure the AI can handle cases where the wire numbering wraps up.
 - Change the comparator to sort hits (Raffaella's recommendation)
- Merge these modifications in the development branch:
<https://github.com/JeffersonLab/coatjava/commits/development/>
- Check that it doesn't change the outputs

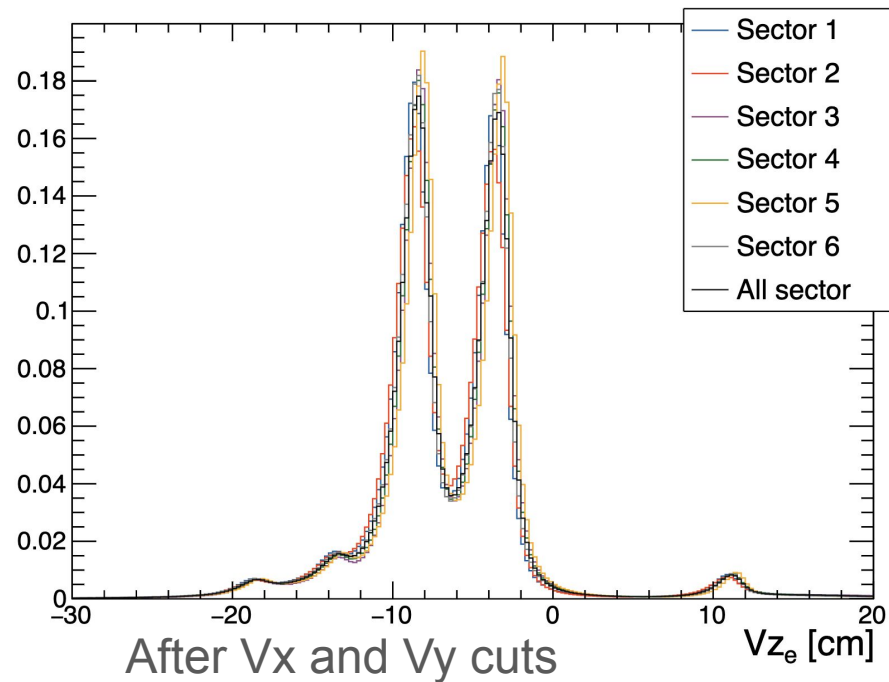
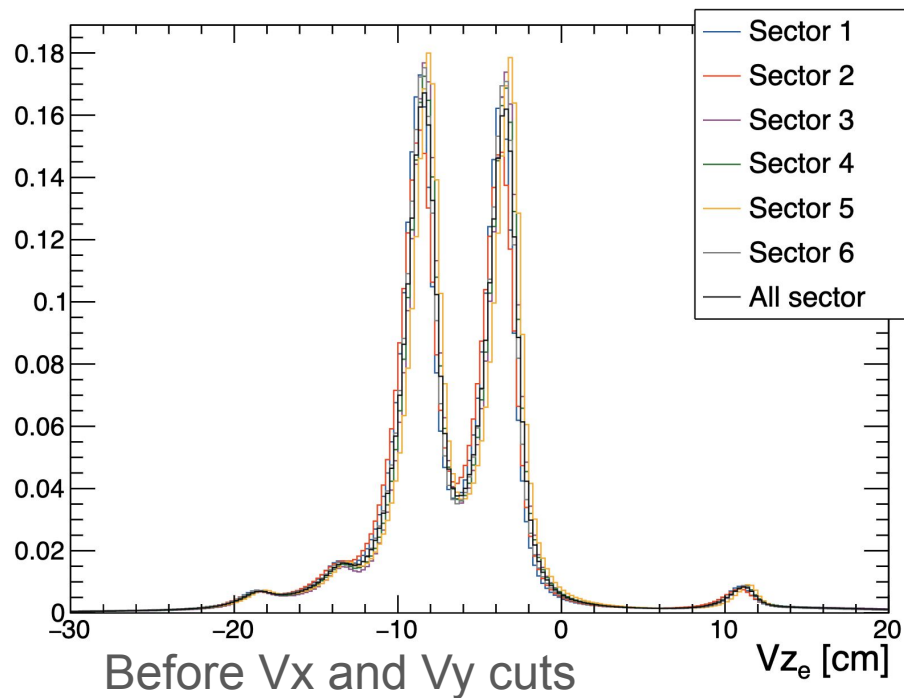
Sector information can come from the REC::Traj bank or the phi information for the electron

```
inline auto determine_sector(double phi) -> int {  
    phi = warp_neg_pos_pi(phi);  
    if(phi < 30 && phi >= -30){ return 1;}  
    else if(phi < 90 && phi >= 30){ return 2;}  
    else if(phi < 150 && phi >= 90){ return 3;}  
    else if(phi >= 150 || phi < -150){ return 4;}  
    else if(phi < -90 && phi >= -150){ return 5;}  
    else if(phi < -30 && phi >= -90){ return 6;}  
    return -1;  
}
```



Vertex study: CxC target

Use Pass0v7 data, select event with a trigger electron with $\text{pid}=11$, $\text{status}<0$, $-5 < \chi^2_{\text{pid}} < 5$
Charged pions identified with $\text{pid}=211$ and $\text{pid}=-211$ and $-10 < \chi^2_{\text{pid}} < 10$ or charged pions.



Electron:

Before:

sector1: 19746436, sector2: 19370913, sector3: 19488850, sector4: 19050576, sector5: 19127145, sector6: 18247564

After:

sector1: 18670368, sector2: 17650148, sector3: 18393219, sector4: 17818830, sector5: 17554660, sector6: 17221795

Keep:

sector1: 94.5%, sector2: 91.1%, sector3: 94.3%, sector4: 93.5%, sector5: 91.7%, sector6: 94.4%

These cuts remove between 5 and 10%.

Pi minus:

before sector1: 3720032, sector2: 3736311, sector3: 3329376, sector4: 3229297, sector5: 3479976, sector6: 3569019

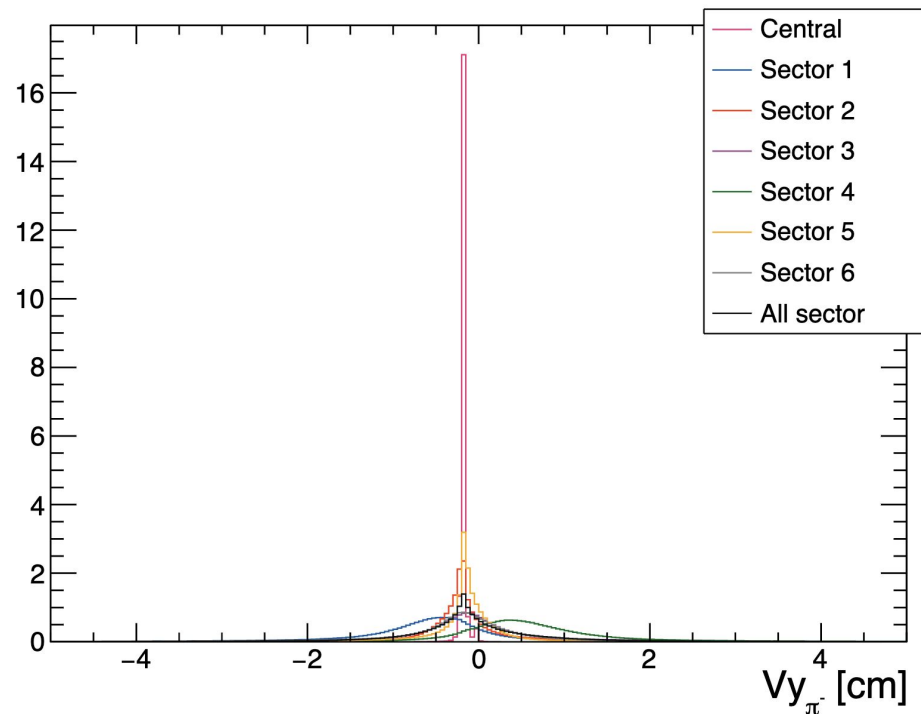
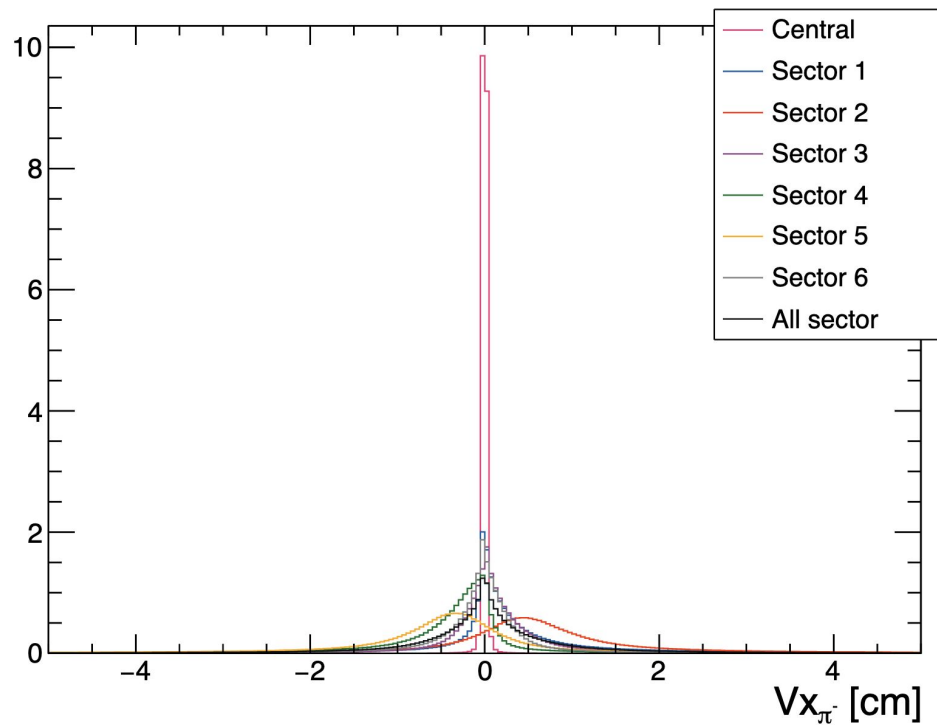
after sector1: 3244445, sector2: 3190361, sector3: 2915273, sector4: 2794007, sector5: 3020196, sector6: 3131147

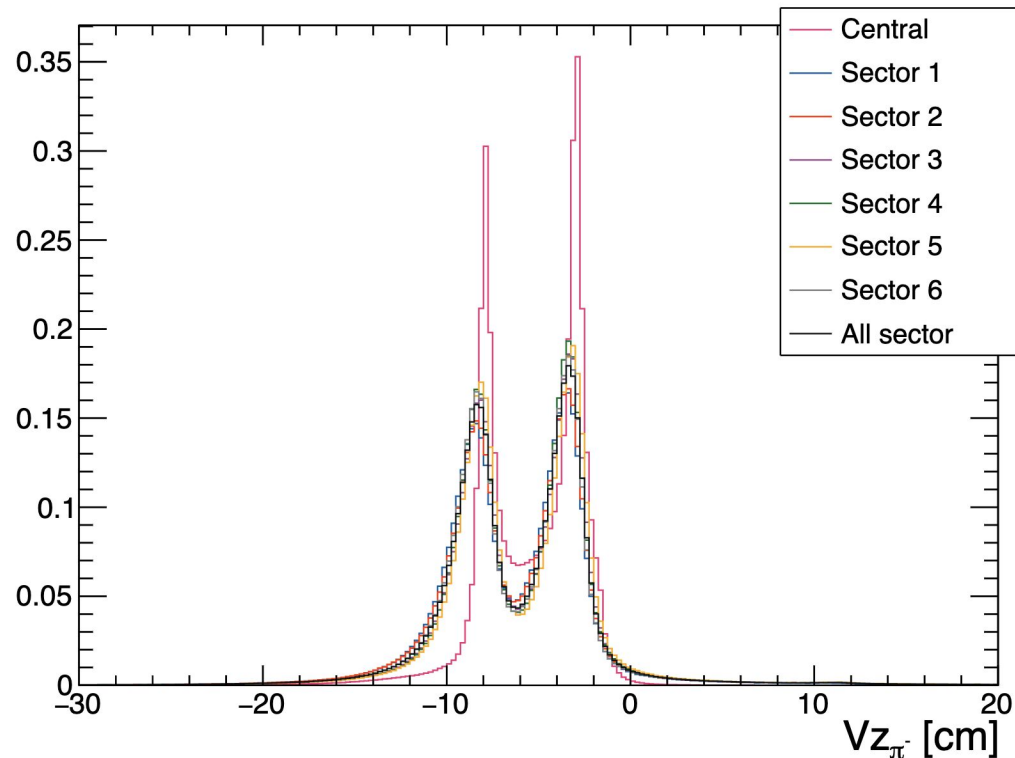
sector1: 87.2%, sector2: 85.4%, sector3: 87.6%, sector4: 86.5%, sector5: 86.8%, sector6: 87.7%

Pi plus:

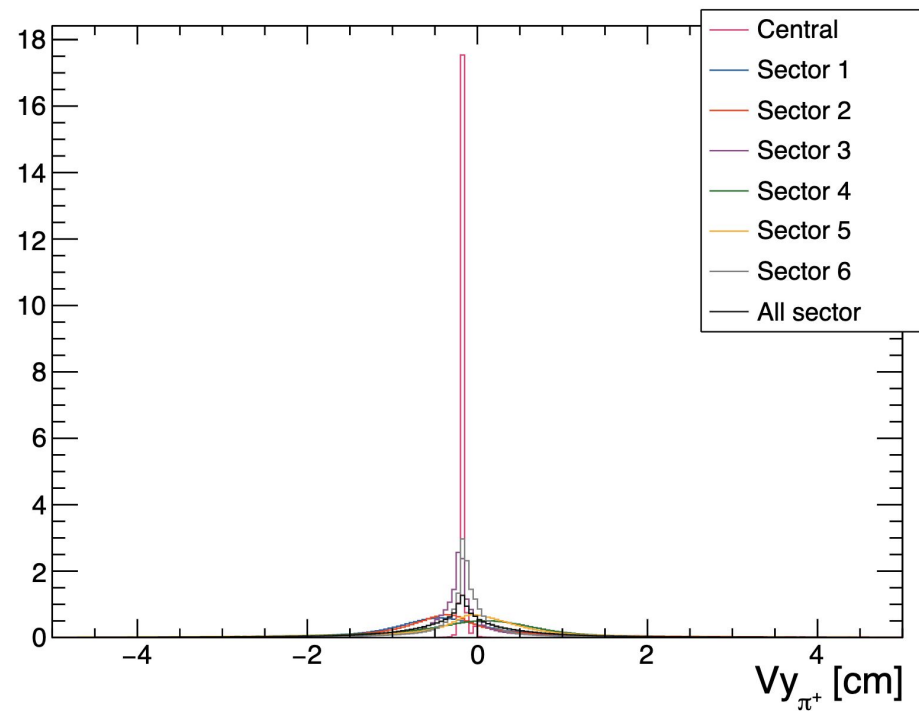
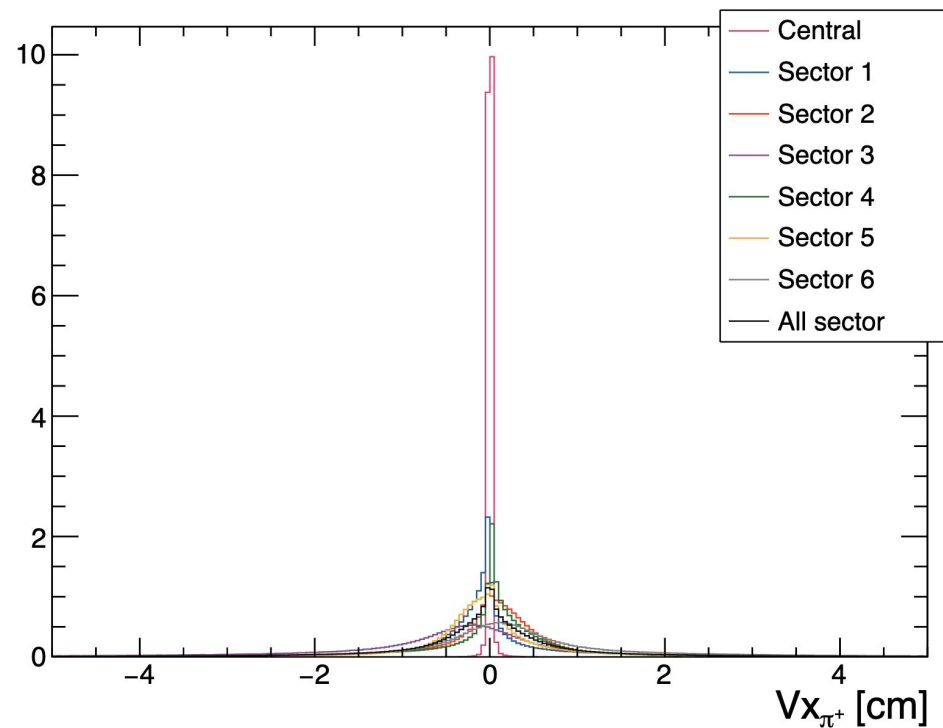
before sector1: 2095371, sector2: 2014860, sector3: 2026940, sector4: 2052339, sector5: 2086435, sector6: 2068615

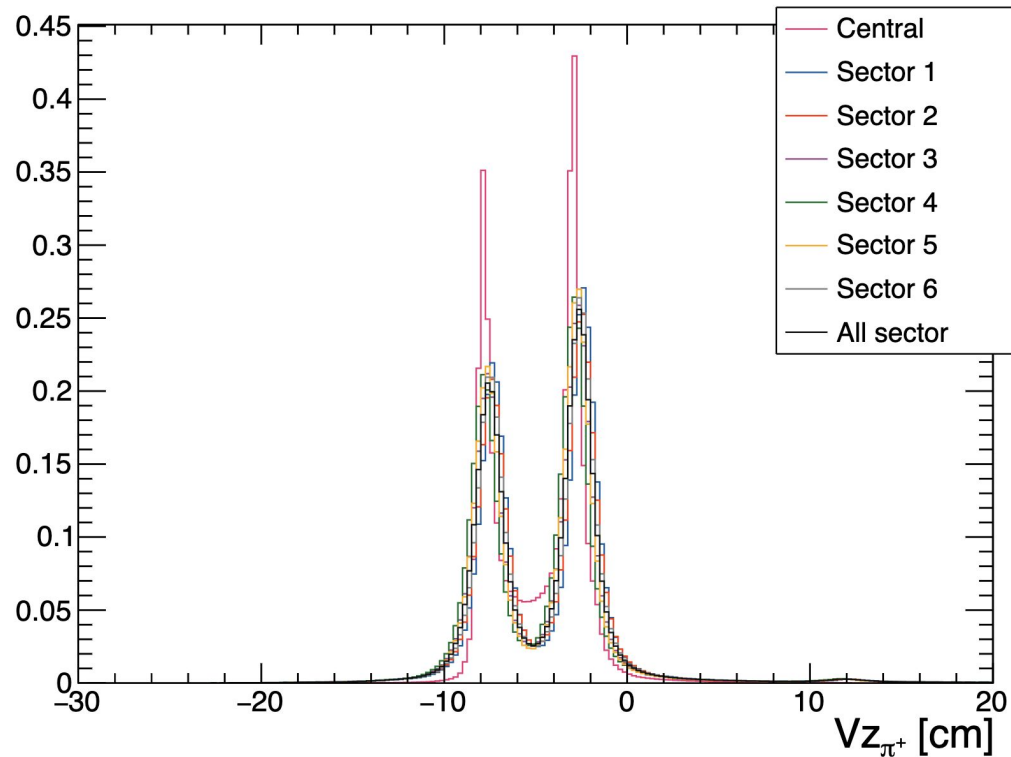
after sector1: 1650244, sector2: 1569280, sector3: 1609778, sector4: 1598219, sector5: 1634720, sector6: 1652999





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- AI for ALERT:
 - Update DJL version from 0.30.0 to 0.31.1
 - Redo the efficiency/purity analysis using luminosity simulation to have a better background (I have generate 1k event to test)
 - Comparison between conventional and AI track finding
- RG-D analysis:
 - Cross check with Matt the V_x , V_y and V_z mean and sigma for each targets
 - Cross check for the nuclear transparency vs Q^2
 - Work on K^0 analysis, test with cuts on z_h and t
 - Work on π^0 analysis, check if I can find them.
- Florian give me a comment about the DC occupancy and FT dose note, Raffaella tell me it is fine.